

Erratum — Three Essays on Factor Models

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How to use this document. Errors are listed by paper and by severity. Fix CRITICAL issues before any submission or final deposit. MAJOR issues affect meaning and should be corrected before the defense copy is printed. MINOR issues (typos, punctuation, duplicates) can be batched and corrected last.

Summary

Category	Count
Total errors	59
Critical (technical / mathematical)	6
Major (grammar / missing words affecting meaning)	22
Minor (typo / punctuation / duplicate word / style)	31

Critical issues at a glance:

- Issue 6 — c_t vs. c : threshold written as time-varying in one equation (Paper 1)
 - Issue 22 — $\hat{\gamma}$ reported as 0.955 in text, 0.995 in figure caption (Paper 1)
 - Issue 46 — Pervasiveness condition sums to m_r instead of N_r (Paper 3)
 - Issue 47 — ξ subscripts conflate factor-dimension and group-index notation (Paper 3)
 - Issue 50 — SVD dimensions: $\mathbf{V}$ described with wrong dimensions (Paper 3)
 - Issue 54 — Portuguese editorial comment left in active LaTeX source (Paper 3)
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Paper 1: Smooth Transition Factor Model

#	Location	Type	Found	Correction
1	Chapters/1introduction.tex, line 11	Typo	"the typical converge rate"	"the typical convergence rate"
2	Chapters/1introduction.tex, line 13	Grammar	"the need of two distinct sets of factors ... in pre and post break subsamples"	"the need for two distinct sets ... in pre- and post-break subsamples" (wrong preposition; missing hyphens in compound modifier)
3	Chapters/1introduction.tex, line 15	Grammar	"the proposal of linearity test against"	"the proposal of a linearity test against" (missing article)

#	Location	Type	Found	Correction
4	Chapters/1introduction, line 19	Grammar,	"Although, the non-parametric approach"	"Although the nonparametric approach" (spurious comma after "Although")
5	Chapters/1introduction, line 21	Grammar,	"section 8 present the empirical application"	"section 8 presents the empirical application" (subject-verb agreement)
6	Chapters/2representation, line 25	CRITICAL — notation	"values of s_t higher than c_t "	"values of s_t higher than c " (threshold is a constant c , not time-varying c_t ; the subscript t is incorrect and implies a moving threshold)
7	Chapters/3estimation, line 23	Grammar	"A common recommendation in literature"	"A common recommendation in the literature" (missing article)
8	Chapters/3estimation, line 55	Grammar	"associated to its largest r eigenvalues"	"associated with its largest r eigenvalues" (wrong preposition)
9	Chapters/3estimation, line 75	Grammar	"convergence to a local minima"	"convergence to a local minimum " ("minima" is plural; context needs singular)
10	Chapters/7finalremarks, line 3	Grammar,	"we follow \{\}citet{chen2014detecting}citet{chen2014detecting} regression-based LM tests to the smooth transition context"	"we adapt \{\}citet{chen2014detecting}citet{chen2014detecting} regression-based LM tests to the smooth transition context" (missing possessive; "follow ... tests to" is ungrammatical)
11	Chapters/7finalremarks, line 5	Missing word	"A smooth transition factor model with provides a good fit"	"A smooth transition factor model with three factors provides a good fit" (number of factors omitted after "with")
12	Chapters/7finalremarks, line 5	Grammar,	"to contribute empirical analysis and forecasting"	"to contribute to empirical analysis and forecasting" (missing preposition)

#	Location	Type	Found	Correction
13	Chapters/5montecarlo.tex, line 45	Notation	In commented block: " $\beta_{f1} = \beta_{e2} = 0.8$ "	Mismatched subscripts — left side uses index 1, right side uses index 2. Correct to " $\beta_{f2} = \beta_{e2} = 0.8$ "
14	Chapters/5montecarlo.tex, line 84	Grammar	Sentence starting "Let ... , the innovations are distributed"	Dangling participle / comma splice. Split into two sentences: "Let The innovations are distributed as ..."
15	Chapters/5montecarlo.tex, line 99	Grammar	"sampled from an uniform distribution"	"sampled from a uniform distribution" (article "a" before consonant sound)
16	Chapters/5montecarlo.tex, line 122	Punctuation	"we simulate a linear factor model with $r = 2$ " (no full stop before equation)	Add full stop: "...with $r = 2$."
17	Chapters/5montecarlo.tex, line 137	Grammar	"does not seems to be very impactful"	"does not seem to be very impactful" (incorrect verb form after "does not")
18	Chapters/5montecarlo.tex, line 141	Missing word	"the power of the test notably higher"	"the power of the test is notably higher" (missing verb)
19	Chapters/6empiricalapplications.tex, line 7	Spelling	"variables with that are available"	"variables that are available" (remove "with")
20	Chapters/6empiricalapplications.tex, line 11	Grammar	"which has a now a long tradition"	"which now has a long tradition" (duplicate "a"; wrong word order)
21	Chapters/6empiricalapplications.tex, line 15	Grammar	"the linear IC_1 ... suggests $\tilde{r} = 6$, which is consistent with the idea that capturing regime-switching dynamics typically requires roughly twice as many linear factors"	The logic is inverted: the <i>linear</i> criterion over-counts because it cannot distinguish regime shifts from additional factors. Rephrase: "... $\tilde{r} = 6$, consistent with the known result that ignoring regime shifts inflates the estimated number of factors."

#	Location	Type	Found	Correction
22	Chapters/6empiricalApplications.tex, line 21 vs. figure caption	CRITICAL: inconsistency	$\hat{\gamma} = 0.955$ in text; $\hat{\gamma} = 0.995$ in figure note	One of these is a transcription error. Verify against estimation output and correct both places to the same value.
23	Chapters/6empiricalApplications.tex, table notes	Application	Table header abbreviation "Dur. Cons." vs. note "Dur. Goods denotes Durable Consumer Goods"	Inconsistent abbreviation. Choose one form and use it consistently in both header and note.

Paper 2: Financial Cycle Measurement

#	Location	Type	Found	Correction
24	1. INTRO.tex, line 5	Citation style	Inline citations used as parenthetical: should use <code>\{citep\}</code> not <code>\{citet\}</code>	Change <code>\{citet{bernanke1999financial}</code> to <code>\{citep{bernanke1999financial}</code> , <code>\{citet{kiyotaki1997credit}</code> to <code>\{citep{kiyotaki1997credit}</code> , <code>\{citet{he2013intermediary}</code> to <code>\{citep{he2013intermediary}</code> where not used as grammatical subjects
25	1. INTRO.tex, line 7	Duplicate citation	<code>\{citet{menden2017di}</code> appears twice in same sentence	Remove the duplicate citation key
26	1. INTRO.tex, line 9	Wrong word	"useful to informal macroprudential policy"	"useful to inform macroprudential policy" ("informal" → "inform")
27	1. INTRO.tex, line 9	Grammar	"the proposed measures exhibits useful"	"the proposed measures exhibit useful" (subject-verb agreement: plural "measures")
28	2. DATA.tex, line 25	Duplicate word	"this transformation is more more suitable"	"this transformation is more suitable" (remove duplicate "more")
29	2. DATA.tex, line 39	Wrong word	"To avoid conflating low-term movements"	"To avoid conflating low-frequency movements" ("low-term" is not standard)
30	3. METHODOLOGY.tex, line 33	Grammar	"a hybrid approach that combine"	"a hybrid approach that combines " (subject-verb agreement)

#	Location	Type	Found	Correction
31	3. <code>METHODOLOGY.tex</code> , line 96	Citation note	EM algorithm cited as <code>\{\}\cit{watson1983alterna}</code>	The standard reference for EM in state-space models is <code>\{\}\cit{shumway1982approach}</code> ; for the general EM it is <code>\{\}\cit{dempster1977maximum}</code> . Verify the author's intended reference.
32	4. <code>RESULTS.tex</code> , line 18	Grammar	"the financial cycle stronger is more closely related to credit"	"the financial cycle is more strongly related to credit" (misplaced modifier)
33	4. <code>RESULTS.tex</code> , line 81	Grammar	"have played an significant role"	"have played a significant role" (article "a" before consonant "s")
34	4. <code>RESULTS.tex</code> , line 92	Grammar	"neither elevated or subdued"	"neither elevated nor subdued" (correlative conjunction: "neither ... nor")
35	4. <code>RESULTS.tex</code> , line 131	Duplicate word	"It serves serves as the recommended"	"It serves as the recommended" (remove duplicate)
36	4. <code>RESULTS.tex</code> , line 133	Wrong phrase	"the area under the operating receiver curve (AUROC)"	"the area under the receiver operating characteristic curve (AUROC)" (standard terminology; correct order)
37	4. <code>RESULTS.tex</code> , line 133	Grammar	"a global measure of accuracy and plot relates"	"a global measure of accuracy and plots " (wrong verb form; "AUROC ... plots")
38	4. <code>RESULTS.tex</code> , line 147	Duplicate word	"the policymaker is is less averse"	"the policymaker is less averse" (remove duplicate "is")
39	4. <code>RESULTS.tex</code> , line 147	Grammar	"we consider balance preferences"	"we consider balanced preferences" ("balance" → "balanced")

Paper 3: Non-stationary DFM with Threshold Cointegration

#	Location	Type	Found	Corro
40	<code>main.tex</code> , line 47	Grammar	"group r-specific variable"	"group variable as "the belong, r"
41	<code>main.tex</code> , line 61	Notation inconsistency	RHS of equation: $\Lambda^* \mathbf{F}_{r,t}^*$ lacks subscript r on Λ^* while $\mathbf{\Gamma}_r^*$ retains it	Change consist
42	<code>main.tex</code> , line 71	Grammar	"normalisations of the factors space"	"norm the (remo plural
43	<code>main.tex</code> , line 71	Grammar	"used as the principle for the estimation"	"used rion tion" "crite "objec after
44	<code>main.tex</code> , line 75	Punctuation	"Since, differencing removes persistent components"	"Since remov tent (remo ous "Since Summ should ber of r), no of factor
45	<code>main.tex</code> , Assumption 2b	CRITICAL — technical	Pervasiveness condition: $\frac{1}{m_r} \sum_{i=1}^{m_r} \gamma_{r,i} \gamma'_{r,i} \rightarrow \Sigma_{\gamma_r}$	$\frac{1}{N_r} \sum$ Re-inc labels or si separa index mensi
46	<code>main.tex</code> , Assumption 3c	CRITICAL — notation	$\mathbb{E}(\xi_{m_0,it} \xi_{m_r,jt})$ — subscripts m_0, m_r conflate factor-dimension parameters with unit/group indices	labels or si separa index mensi
47	<code>main.tex</code> , line 181	Grammar	"allows to effectively separate"	"allow tively "allow separa
48	<code>main.tex</code> , line 213	Punctuation	"Stack ... $\mathbf{x}_{r,t}$ $r = 1, \dots R$ "	"Stack r (add \{\}1d

#	Location	Type	Found	Correction
49	main.tex, line 217	CRITICAL — technical	SVD description: " $\mathbf{V}$ is $T \times N$ "	For $\mathbf{V}$ is or N (thin) dimer corres not $\mathbf{V}$ dimer to m varian
50	main.tex, line 234	Grammar	"Let $\mathcal{F}_t$ denote the stacked vector of all group-specific factors $Q = \sum_{r=1}^R m_r$ "	Add $\sum_{r=1}^R$ mensi tion c it equ
51	main.tex, line 267	Grammar	"particularly useful to model asymmetries in the adjustment towards equilibrium"	"parti to : metri adjust equili one use throu
52	main.tex, line 303	Notation	In equation: " Φ_{2i} " (no comma between subscripts)	" $\Phi_{2,i}$ " for co $\Phi_{j,i}$ el
53	main.tex, line 539	CRITICAL — editorial note left in source	"Isto pode apagar-se?"	This comm this b inside source before for s depo
54	main.tex, line 421	Technical	In Remark: " $\mathbf{x}, \mathbf{y} \in \mathbb{R}^R$ " — dimension is R	The c trix d not to $\mathbb{R}$ the d above

Thesis Introduction and Conclusion

#	Location	Type	Found	Correction
55	thesis_intro.tex, line 17	Duplicate citation	<code>\citep{su2017time}</code> — same key cited twice	Remove the duplicate <code>su2017time</code> from the citation list

#	Location	Type	Found	Correction
56	conclusion.tex, line 5	Style	"With the advancement of data collection capability"	"With advances in data collection " (more natural phrasing for academic conclusion)
57	conclusion.tex, line 7	Style	"often predicted by economic theory and observed in practice"	"often predicted by economic theory and frequently observed in practice" (minor formality improvement)

Critical Issues — Detail Notes

Issue 6 — Threshold written as c_t instead of c (Paper 1)

In `2representation.tex` line 25, the location parameter of the logistic transition function is written as c_t (with a time subscript), but the model defines c as a constant parameter. This is not just a notation inconsistency — it implies the threshold moves over time, which would change the model interpretation entirely. Verify whether this is a typo or an intentional generalisation (if intentional, the estimation section and Monte Carlo must also reflect a time-varying threshold).

Issue 22 — $\hat{\gamma}$ discrepancy (Paper 1)

The estimated smoothness parameter is reported as $\hat{\gamma} = 0.955$ in the text (line 21 of `6empiricalapplication.tex`) and $\hat{\gamma} = 0.995$ in the figure caption for the transition function plot. These two numbers differ enough ($\sim 4\%$) that one must be a transcription error. Check the estimation output directly and correct both occurrences to the same value.

Issue 45 — Pervasiveness condition (Paper 3)

In Assumption 2(b), the pervasiveness condition on global loadings $\gamma_{r,i}$ sums over $i = 1, \dots, m_r$ (the number of group-specific factors), but the correct index should run over $i = 1, \dots, N_r$ (the number of cross-sectional units in group r). The standard pervasiveness condition in factor models (Chamberlain–Rothschild, Bai–Ng) sums over units, not over the factor dimension. This is a technical inaccuracy in the assumption statement.

Issue 46 — Conflated notation in Assumption 3 (Paper 3)

The subscripts m_0 (number of global factors) and m_r (number of group- r factors) are used on the idiosyncratic error ξ in Assumption 3(c), making it appear as if the error is indexed by the factor-dimension parameters rather than by the group and unit indices. Re-index with distinct symbols (e.g., r_1, r_2 for groups; i, j for units within groups) to eliminate the ambiguity.

Issue 49 — SVD dimension mismatch (Paper 3)

The algorithm description states " $\mathbf{V}$ is $T \times N$ ", but in the standard SVD $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}'$, $\mathbf{V}$ is always $N \times K$ (where $K = \min(T, N)$ for thin SVD or $K = N$ for full SVD). The $T \times N$ dimension corresponds to $\mathbf{V}'$ (the transpose), not $\mathbf{V}$ itself. Confirm which convention is being used and correct the stated dimensions accordingly.

Issue 53 — Portuguese editorial comment in source (Paper 3)

The phrase "Isto pode apagar-se?" (Portuguese: "Can this be deleted?") appears at line 539 of `main.tex` inside the active LaTeX body, not within a comment. It will appear verbatim in the typeset output. **This must be removed before the thesis is deposited or printed.**